Yaoyuan Zhou

My Website amiao1120@gmail.com GitHub

Education

Beijing Normal University *Sept 2023 – Jul 2027*
BS in Educational Technology (major) **Average Score: 87.27 (G:3.7/4.0)**
BS in Artificial Intelligence (minor)
Hong Kong Polytechnic University *Sept 2025 – Jan 2026*
Exchange Student - BSc(Hons) in Artificial Intelligence and Information Engineering

Awards

Second-Class Jingshi Scholarship, First-Class Merit-Based Scholarship
The NTU National Institute of Education (NIE) Singapore Undergraduate Online Scholarship

Research Interest

Core Research Interest: My research focuses on AI-supported pedagogical design and intelligent educational tool development, particularly on introducing novice learners to AI and fostering their foundational AI literacy.
Core Research Goal: From the standpoint of Education, leveraging the potential of Technology in pedagogical practices to explore pathways towards harmonious coexistence between humanity and the world.

Publications

Journal

An Active Instructional Approach Based on SAMR Framework to Assist Undergraduate Freshmen in Integrating AIGC into Learning: An Exploratory
The Internet and Higher Education**(SSCI Q1, Third Author)** Link
Juan Wu*, Jingwen Pan, **Yaoyuan Zhou**, Mengyu Liu, Yanling Li, Ronghuai Huang,

Exploring the Second-Level Digital Divide in Elementary Science Classrooms: A Video-Based Comparative Analysis
International Journal of Science and Mathematics Education**(SSCI Q1, Second Author)** Link
Yuze He, **Yaoyuan Zhou**, Juan Lai, Fengrui Xie, Yonghe Zheng, Jie Yang*

Exploring the Effects of Generative AI Prompting Instruction on Undergraduate Students' Creative Problem Solving: From Learning to Using in a Classroom-Based Social Experiment
Smart Learning Environment (**SCI Q1, Third Author**,Under Review)
Juan Wu*, Jingwen Pan, **Yaoyuan Zhou**, Mengyu Liu, Ronghuai Huang

Conference

Exploring the Pattern and Influence Factors in the Correlation Between Learning Engagement and Academic Achievement in Online Learning
Global Chinese Conference on Computers in Education, GCCCE 2026 **(First Author)**
Yaoyuan Zhou, Yuhan Gai, Qingyang Liu, Jiamin Ren

Power of Query——Analysis of Critical Thinking Trajectories Based GAI-Assisted Learning
Global Chinese Conference on Computers in Education, GCCCE 2025 **(First Author)** Link
Yaoyuan Zhou, Juan Wu*

A Typological Study of Learning Engagement and Academic Achievement in University Online Classes :An Epistemic Network Analysis Based on Bloom's Taxonomy
Research Association of Learning Science Conference, RALS 2025 **(First Author)**
Yaoyuan Zhou, Yuhan Gai, Qingyang Liu, Jiamin Ren

Project

Exploring the Effects of GAI Prompting on Undergraduate Creative Problem Solving: A Task-Driven Experimental Study ***Person in Charge*** *May 2024 - Dec 2025*
Municipal-level scientific research project

- Led the end-to-end design and implementation of a classroom AI-agent platform supporting real-time student–GAI interaction.

- Designed a prompt-engineering curriculum and analyzed students' chatbot dialogues using Epistemic Network Analysis (ENA) to examine their critical and creative thinking.

The scientific and engineering education discipline community of Beijing Normal University ***Project Intern***
Oct 2023 – Jun 2025
Research Institute of Science Education

- Designed a 12-hour aviation-themed STEM curriculum grounded in Project-Based Learning (PBL), developing hands-on learning tasks and co-authoring the accompanying textbook materials.
- Collaborated with 20+ frontline STEAM educators, serving between the research team and teachers to translate classroom feedback into iterative curriculum revisions and coordinate textbook development.

A systematic review of intelligent tutoring systems based on Generative Artificial Intelligence ***Person in Charge***
Aug 2025 - Now
The Education University of Hong Kong - Hong Kong, China

- Lead a systematic review and meta-analysis examining how generative AI reshapes classical ITS architectures and interaction mechanisms, covering protocol development, literature screening, data extraction, and quantitative synthesis.

Research on the Systematic Thinking of Artificial Intelligence-Assisted Reading in Chemistry Teaching for Middle School Students ***Person in Charge***
March 2026 - Now
Beijing Normal University & Beijing No.4 High School(BHSF)

- Lead the research design, curriculum development, agent prototyping, and classroom implementation of an AI-supported chemistry intervention aimed at fostering students' systems thinking.

Enhancing Education: A Human-Centred Generative AI Pipeline for Academic Excellence ***Research Assistant***
Jul 2026 - Oct 2026
Mitacs Globalink Research Internship Program (University of Regina) - Canada

- Suport the design and evaluation of an LLM-based educational system that adapts content to student needs, focusing on intelligent assessment, curriculum peronalization, and real-time feedback mechanisms.
- Conducted quantitative research involving benchmarking, performance analysis, and statistical modeling to validate system efficacy, integrating findings into iterative development cycles.

Work Experience

Project Intern - The UNESCO International Research and Training Centre for Rural Education (INRULED) *Feb 2025 – Aug 2025*
Information and Communication Technology (ICT) in Education for Rural Development Department

- Facilitated the joint SEAMEO STEM-ED and UNESCO INRULED "STEM Project" by conducting background research on rural STEM education, drafting cooperation frameworks and evaluation criteria.
- Assisted 5 large-scale international events, managing concept notes, round-table session design, foreign expert liaison, and on-site multimedia in high-pressure, multilingual settings.
- Conducted field surveys across Chongqing, Jiangsu, and other regions to assess the current implementation status of ICT education.

Online & Offline Teacher Volunteer - Teaching Digital Skills and SDGs to Primary School Students in Rural China (UNDP) *Jan 2024 - Jan 2025*

- Served as an AI literacy instructor at the 2024 HER Digital Future Camp, delivered 20 hours of instruction to primary school students and promoting girls' participation in STEM through hybrid learning experiences.
- Developed introductory AI lessons based on prompt-driven human–AI interaction, fostering students' critical thinking, problem-solving abilities, and early self-regulated learning skills.

Special Skills

Languages: English(IELTS 7.0),Chinese(Native Speaker), Cantonese(Beginner)
AI Capacity: PyTorch, Transformers, scikit-learn, pandas, NumPy; BERT and LLM-based text classification
Research Method: ENA, process mining, lag sequential analysis, systematic review and meta-analysis